

Matthew Molineaux, Ph.D.

AI SYSTEMS ARCHITECT · AUTONOMY & AGENT EVALUATION

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scholar.google.com/citations?user=9Zru6S0AAAAJ

PROFILE

AI systems architect. Twenty years building autonomous systems that keep working after the world stops matching the model. Delivered a non-generative architecture that met or exceeded targets in all four independent evaluations of a \$6M DARPA program, beating the competing generative-AI team every round. Now targeting architecture and technical leadership in commercial autonomy.

TECHNICAL FOCUS

Autonomous agents — Agent architectures; goal reasoning and goal formulation; hierarchical and temporal planning; causal explanation and diagnosis; execution monitoring and plan repair; plan and activity recognition; multi-agent systems; human-agent teaming; constraint and combinatorial optimization; answer set programming; verified tool use; computer-use agents

Learning and evaluation — Reinforcement learning; environment and action model learning; experiment design and metric definition; independent and adversarial evaluation; human-alignment measurement; open-world robustness under distribution shift

Leadership — Architecture and interface specifications other engineers build from; technical roadmap and design review across concurrent programs; delivery through LLM-assisted implementation and junior staff; budget ownership

Application domains — Autonomous underwater and ground vehicles; clinical triage and decision support; aircraft and fleet maintenance; public-health resource allocation; adversarial simulation

EXPERIENCE

Independent AI Product Development

Feb 2026 – Present

Founder & Technical Lead

Washington Township, OH

Telotrope Agents, Inc. (Jun 2026–Present) · Delegation, Inc., co-founder (Feb–Aug 2026)

- Built an agent that walks a website the way ten different customers would, each with their own goal, budget, and patience, and reports where they gave up. Output: conversion rate, which questions the site answered, which elements no visitor ever found. Run across 50 sites, every report reviewed for fidelity to real user experience.
- Made it admit what it could not see — widgets it failed to operate, pages it could read but not enter, actions never reached. Coverage gaps get reported, not papered over.
- Designed a goal-reasoning system that executes business processes, extending goal formulation to commercial workflows, and an interface designer that turns natural-language direction into UI screens.
- Developed verified tool-use planning: the agent builds and checks a plan over available tools before acting, so a failure surfaces as a plan violation instead of a confident wrong answer. Also prototyped inference of a person's goals and emotional state from observed behavior.
- One architecture throughout: explicit models and instrumentation hold the guarantees; generation is fenced into the corners where invention cannot pass silently.

Parallax Advanced Research

Feb 2018 – Apr 2026

Director of AI and Autonomy

Beavercreek, OH

- Principal Investigator, \$6M program awarded. Designed the Trustworthy Algorithmic Delegate (TAD), which makes triage decisions matching a specific clinician's judgment rather than a population average, built on case-based reasoning, Monte Carlo simulation, and Bayesian diagnosis. Four independent evaluations, four times meeting or exceeding targets, on schedule — beating the competing generative-AI team every round with a system that does not generate.

- Ran a 10-person group across AI, machine learning, neuromorphic computing, and cognitive science: technical direction, architecture, and staffing for a portfolio of concurrent programs. Hired ~10 technical staff; graduated 3 PhD and 2 MS students.
- Brought in tens of millions over eight years by winning the same customers again and again, owning each bid from problem framing through architecture, pricing, and staffing.
- Built agents for worlds that change out from under them: environment models learned by self-directed exploration, plans repaired mid-execution, detection of conditions its own model does not cover, and a framework that transforms environments to measure how far an agent degrades.

Knexus Research Corporation

Apr 2007 – Jan 2018

Senior Computer Scientist

National Harbor, MD

- Designed ARTUE (Autonomous Response to Unexpected Events), a domain-independent agent that decides what goal to pursue when something surprising happens: explain the discrepancy first, then form a goal from the explanation. Outperformed offline planning and replanning baselines; independent groups still analyze the architecture and build alternatives against it more than a decade later. Extended its hierarchical task network planner with nonlinear continuous effects, letting it reason about physics rather than discrete state.
- Invented DiscoverHistory: reasoning backward from partial observations to what must have happened, over mixed discrete and continuous data. An agent using it learned an unseen environment dynamic in five examples and raised goal achievement 50% over an agent that never asks why.
- Designed three generations of replanning-and-execution systems for physical operations: CARPE (maritime survey and clearance), ADRUS (anomaly detection and recovery for unmanned systems), and scheduling and repair for autonomous underwater vehicles. One thesis across all three: when a mission breaks mid-execution, repair the plan instead of restarting it.
- Designed ASM (Autonomous Squad Member), an agent built to act as a team member alongside people, taking natural-language direction; developed and evaluated in simulation.
- Advised NRL and DARPA on how to measure explainability in autonomous agents.

ITT Industries

Dec 2003 – Apr 2007

Research Scientist

Alexandria, VA

Full-time under contract to the Naval Research Laboratory, Washington, DC

- Designed and built TIELT, the integration layer for a DARPA program: one interface that let learning agents from 5–10 independent teams run against simulated environments none of them had built, including a commercial game engine (later Rockstar New England). Owned the abstraction and the documentation those teams built against.
- Designed the Case-based Tactician (CaT), which learns to beat opponents in a real-time strategy environment by retrieving and adapting prior engagements. ICCBR Best Paper.
- Designed and ran the technology evaluations for the DARPA Transfer Learning program: defined the metrics, analyzed results across performer teams.

EDUCATION

Ph.D., Computer Science	George Mason University	2017
M.S., Computer Science	DePaul University	2008
B.S., Computer Science	Eckerd College	2001

RECOGNITION AND PUBLICATIONS

DARPA Riser, 2022 · Best Paper: ICCBR 2005, FLAIRS 2008 and 2011 · Best Video, IJCAI 2009

65+ peer-reviewed publications in planning, goal reasoning, explanation, and agent evaluation — [Scholar](#)

Selected languages and tools — C++, Common Lisp, Rust, Python · PDDL, Clingo/ASP, SHOP2, Fast Downward · PyTorch, MCP, local open-weight models · ZeroMQ, MQTT, SOAP, AWS · ROS 2, Gazebo